FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Anaemias & Blood Transfusion

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Nabatte

Anaemia in Ugandan children is usually multifactorial (malaria, iron deficiency, sickle cell disease, HIV, worms, malnutrition), so treat both the Hb and its causes; transfuse by Hb plus clinical severity (Hb ≤4 g/dL, or 4–6 g/dL with danger signs), give 20 mL/kg whole blood or 10 mL/kg packed cells over 3–4 hours, and watch closely for overload and reactions.

Must know before the ward round

1. WHO 2024 Hb cut-offs: anaemia below 10.5 g/dL at 6–23 months, below 11.0 at 24–59 months, below 11.5 at 5–11 years and below 12.0 at 12–14 years. Severe anaemia is below 7 g/dL under 5 years and below 8 g/dL from 5 years.
2. Transfuse (WHO Pocket Book 2013 / UCG 2023) every child with Hb ≤4 g/dL (Hct ≤12%), and a child with Hb 4–6 g/dL who has respiratory distress (acidotic breathing), shock, impaired consciousness, heart failure, dehydration or hyperparasitaemia.
3. Volume: whole blood 20 mL/kg or packed cells 10 mL/kg (UCG: 10–15 mL/kg) over 3–4 hours. Rule of thumb: 8 mL/kg whole blood or 4 mL/kg packed cells raises Hb by about 1 g/dL.
4. Severe acute malnutrition: only 10 mL/kg over 3 hours, give no furosemide routinely, and do not repeat within 4 days.
5. TRACT (Uganda/Malawi, NEJM 2019): in uncomplicated Hb 4–6 g/dL, immediate transfusion gave no survival benefit over monitoring with a trigger. Among afebrile children with complicated severe anaemia, 30 mL/kg cut mortality compared with 20 mL/kg.
6. Start blood within 30 minutes of leaving the fridge and finish within 4 hours. Record vitals before, at 15 minutes, every 30 minutes and at the end. Watch the first 15 minutes most closely.
7. Any fever ≥39 °C, rigors, hypotension, breathlessness, back pain or dark urine during transfusion: STOP, keep the line open with saline, check the patient's ID against the bag, resuscitate, and send the bag and fresh samples to the blood bank.
8. Iron deficiency: microcytic and hypochromic, low ferritin (<12 µg/L under 5 years if no inflammation). Give ferrous sulphate about 3–6 mg/kg/day elemental iron (UCG 2023: 5 mg/kg of the salt 8-hourly, max 200 mg/dose) for 3 months after Hb normalises, plus deworming.
9. SCD prophylaxis (UCG 2023): penicillin V 125 mg 12-hourly from 3 months to 3 years and 250 mg 12-hourly from 3 to 5 years, folic acid 5 mg daily, monthly SP malaria chemoprevention, full UNEPI vaccines including PCV, and hydroxyurea from 9 months.
10. Hydroxyurea starts at 20 mg/kg once daily (UCG 2023; REACH 15–20 mg/kg). Escalate by 5 mg/kg every 8 weeks to the maximum tolerated dose (max 35 mg/kg/day), checking FBC. NOHARM (Uganda) showed it does not increase malaria.
11. SCD with fever ≥38.5 °C is an emergency: blood culture, malaria test, then ceftriaxone 80–100 mg/kg IV within 1 hour, because hyposplenic children die of pneumococcal sepsis.
12. Falling Hb with an enlarging spleen means splenic sequestration (retics high). Falling Hb with retics under 1% means aplastic crisis (parvovirus B19). Both need urgent transfusion. In sequestration give small aliquots (5–10 mL/kg) because the spleen gives cells back.

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In a nutshell

Anaemia means a haemoglobin (Hb) level below the normal range for the child's age. In Uganda it is one of the commonest reasons a child is admitted, and **severe anaemia** is a leading cause of death in children under 5, mostly from **malaria**. It is rarely a single disease: a pale 2-year-old in Kampala may have malaria, iron deficiency, hookworm, HIV and poor diet all at once. That is why we say **"anaemia is a sign, not a diagnosis."**

Two questions drive everything:

1. **Does this child need blood now?** That depends on the **Hb and the clinical picture**: respiratory distress, shock, impaired consciousness, heart failure. The Hb number alone does not decide it.
2. **Why is this child anaemic?** Classify by **MCV** (small, normal or big cells) and **mechanism** (not made, destroyed or lost), then treat every cause you find.

Sickle cell disease (SCD) is the commonest inherited cause and you will see children in crisis on almost every ward round.

> CLINICAL PEARL
> The one thing never to forget: **a child with severe anaemia and deep, laboured breathing is dying of tissue hypoxia and acidosis, not of pneumonia.** Blood (and oxygen) saves that child. Giving fluid boluses can kill them.

Definitions & classification

Hb thresholds by age

The WHO revised its anaemia cut-offs in **March 2024** (*Guideline on haemoglobin cutoffs to define anaemia in individuals and populations*). The Uganda Clinical Guidelines (UCG) 2023 still print the older WHO 2011 table, which used one cut-off (11 g/dL) for all children aged 6–59 months.

Age	Anaemia (WHO 2024)	Mild	Moderate	Severe
6–23 months	**<10.5 g/dL**	9.5–10.4	7.0–9.4	**<7.0**
24–59 months	**<11.0 g/dL**	10.0–10.9	7.0–9.9	**<7.0**
5–11 years	**<11.5 g/dL**	11.0–11.4	8.0–10.9	**<8.0**
12–14 years	**<12.0 g/dL**	11.0–11.9	8.0–10.9	**<8.0**

Newborns are different. Hb is high at birth (>13.5 g/dL) and falls to its lowest point around **8–12 weeks** (*physiological nadir*, often 9.5–11 g/dL). Do not label a healthy 2-month-old with Hb 10 g/dL as iron deficient.

UGANDA CONTEXT

UCG 2023 uses Hb <11 g/dL for 6 months–5 years, with severe <7 g/dL. The **WHO Pocket Book (2013)** uses Hb <9.3 g/dL for "non-severe anaemia needing iron" in children under 6 years. For transfusion decisions every guideline uses the same **≤4 and 4–6 g/dL** bands, so those are the numbers that matter on the ward.

Classification by red cell size (MCV)

Normal MCV rises with age. A rough lower limit is **70 + age in years (fL)** up to about 80 fL.

Type	MCV	Common causes in Uganda
Microcytic	Low	**Iron deficiency** (most common), α- or β-thalassaemia trait, anaemia of chronic disease (late), lead poisoning, sideroblastic (isoniazid)
Normocytic	Normal	**Malaria**, acute blood loss, **sickle cell disease**, HIV, chronic disease (TB, chronic kidney disease), marrow infiltration (**leukaemia**, Burkitt), aplastic anaemia, mixed deficiency
Macrocytic	High	**Folate deficiency** (SCD without folic acid, malnutrition, goat's milk), B12 deficiency (exclusively breastfed infants of vegan/B12-deficient mothers), **zidovudine (AZT)**, hydroxyurea (expected), high reticulocytes, hypothyroidism, Down syndrome

Classification by mechanism

Mechanism	Clue	Examples
Decreased production	Low reticulocytes	Iron/folate/B12 deficiency, HIV, TB, malnutrition, leukaemia, aplastic crisis (parvovirus B19), chronic kidney disease, visceral leishmaniasis (kala-azar)
Increased destruction (haemolysis)	High reticulocytes, jaundice, dark urine, raised bilirubin/LDH	**Malaria**, SCD, **G6PD deficiency** (after primaquine, dapsone, cotrimoxazole, fava beans), autoimmune haemolysis, hypersplenism, blackwater fever
Blood loss	History, melaena, iron deficiency if chronic	**Hookworm**, schistosomiasis (haematuria, *S. mansoni* GI loss), trauma, epistaxis, bleeding disorders, peptic ulcer, Meckel's diverticulum

Epidemiology & Uganda context

- About **half of Ugandan children aged 6–59 months are anaemic** (Uganda Demographic and Health Survey data). Prevalence is highest in the second year of life, when iron needs are high and complementary foods are poor in iron.
- **Malaria** is the biggest driver of **severe** anaemia, especially in high-transmission regions (West Nile, Acholi, Lango, Busoga, Karamoja). Severe malarial anaemia peaks between **6 months and 3 years**.
- **Sickle cell disease**: about **1 in 7 Ugandans carries the sickle gene** (higher in the north and east), and an estimated **15,000–20,000 babies are born with SCD each year**. Without care, most die before 5, mainly from pneumococcal sepsis, malaria and acute anaemia.
- **Blood supply**: the **Uganda Blood Transfusion Service (UBTS)** collects, screens (HIV, hepatitis B and C, syphilis) and distributes all blood through regional blood banks. Blood is **available from HC IV level upwards** (UCG 2023), but stock-outs are common, especially during school holidays when student donors are away. Packed cells come in **paediatric bags of 80–150 mL**. Platelets, FFP and cryoprecipitate are mainly found at regional and national referral hospitals.
- Children with severe anaemia who survive often die or are readmitted in the **6 months after discharge** (seen in the TRACT trial). Post-discharge follow-up and malaria chemoprevention matter.

Pathophysiology

Why severe anaemia kills

Oxygen delivery = cardiac output × arterial O_2 content, and O_2 content depends mainly on Hb. When Hb falls slowly (iron deficiency), the body adapts: the heart pumps more, 2,3-DPG rises and oxygen is released more easily, and plasma volume expands. A child can walk into clinic with Hb 4 g/dL.

When Hb falls **fast** (malaria, sequestration) or the compensation is overwhelmed:

1. Tissues switch to anaerobic metabolism, which causes **lactic acidosis**. The child breathes deeply to blow off CO_2. This is the **"respiratory distress" of severe anaemia**, and it is usually not pneumonia.
2. The heart, already working hard, fails (**high-output heart failure**: tachycardia, gallop, big tender liver).
3. The brain gets too little oxygen, which causes **impaired consciousness** and seizures.

This is why WHO uses **clinical severity**, not Hb alone, to decide on transfusion.

Iron deficiency

Iron stores at birth last about **4–6 months** in a term baby (less in preterm, low-birth-weight and twins, and with early cord clamping). After that, iron must come from food. **Breast milk** iron is well absorbed but small in amount. **Cow's milk** is iron-poor, inhibits iron absorption and can cause occult gut bleeding in infants. **Hookworm** causes chronic loss. Iron deficiency also impairs **neurodevelopment**, so it matters even before the child is very anaemic.

Sickle cell disease

- A single amino-acid change (**valine for glutamic acid at position 6 of the β-globin chain**) makes **HbS**. Children with **HbSS** (sickle cell anaemia) have the most severe disease. HbSC and HbS-β-thalassaemia vary.
- When deoxygenated, HbS **polymerises**. Red cells become rigid and sickled, which causes **haemolysis** (cells live about 10–20 days instead of 120) and **vaso-occlusion** (cells block small vessels and cause ischaemia and pain).
- **Fetal haemoglobin (HbF)** does not sickle. It protects for the first **3–6 months**, which is why dactylitis is often the first symptom at 6–12 months. **Hydroxyurea works by raising HbF**, and it also lowers the white cell count and reduces red cell stickiness.
- The **spleen** is damaged early by repeated infarction. Functional asplenia begins in infancy, so children are vulnerable to **encapsulated bacteria** (pneumococcus, Hib, Salmonella). Before it auto-infarcts, the spleen can suddenly trap blood (**splenic sequestration**).

> **EXAM FAVOURITE**
> Why does SCD present after 6 months? Because **HbF ($\alpha_2\gamma_2$)** protects until the switch to adult **β-chain** production is complete. Hydroxyurea exploits this by switching γ-chain production back on.

Clinical acumen

Focused history

Ask about	Why it matters
Fever, recent malaria, where they live and travelled	Malaria is the commonest cause of acute severe anaemia; also sepsis

Ask about	Why it matters
Speed of onset: sudden weakness, fast breathing, collapse vs months of tiredness	Acute = haemolysis, sequestration, bleeding (decompensates early); chronic = deficiency (well compensated)
Dark/cola urine, yellow eyes	Haemolysis: malaria, G6PD, blackwater fever, SCD
Breathing fast, sweating, poor feeding, swelling	Heart failure, which changes how you transfuse (packed cells, slow, furosemide)
Diet: breastfeeding, age at complementary foods, **cow's/goat's milk** intake, meat/beans/greens, **pica** (eating soil)	Iron or folate deficiency; pica is a classic iron-deficiency sign
Worms, blood in stool or urine, bare feet, swimming in lakes	Hookworm; schistosomiasis
Bleeding: nose, gums, bruising, heavy periods in adolescents	Bleeding disorder, leukaemia, aplastic anaemia
Bone pain, swollen hands/feet, previous transfusions, family history of SCD	SCD; also leukaemia (bone pain)
Drugs: cotrimoxazole, AZT, primaquine, dapsone, herbal medicines, antiepileptics	Marrow suppression, G6PD haemolysis, folate antagonism
HIV status of child and mother, TB contact, weight loss, night sweats	HIV, TB, malignancy
Immunisation (UNEPI), growth, development	Missed PCV/Hib in SCD; malnutrition; iron deficiency harms development
Previous transfusions (number, last date, reactions)	Alloantibodies, iron overload, transfusion-transmitted infection

Examination

- **Pallor**: look at the **palms** (IMCI "severe palmar pallor"), conjunctivae, tongue and nail beds. Palmar pallor is the most practical sign in dark skin.
- **Signs of decompensation**: respiratory rate and pattern (**deep acidotic breathing**), heart rate, **capillary refill**, conscious level (AVPU), gallop rhythm, **liver size**, basal crackles, oedema.
- **Jaundice** (haemolysis), **splenomegaly** (malaria, SCD sequestration or hypersplenism, leukaemia, kala-azar, HbSC), hepatomegaly.
- **Growth**: weight-for-height Z-score, MUAC, bilateral pitting oedema (SAM changes transfusion rules).
- **Bleeding signs**: petechiae, bruises, gum bleeding, which suggest marrow failure or leukaemia.
- **Lymph nodes, abdominal masses, bone tenderness**: leukaemia, lymphoma.
- **Frontal bossing, maxillary hypertrophy, dactylitis, leg ulcers**: chronic haemolysis or SCD.
- **Koilonychia, angular stomatitis, glossitis**: iron deficiency (late signs).

Red flags

RED FLAG

- **Hb ≤4 g/dL**, or Hb 4–6 g/dL with **respiratory distress, shock, impaired consciousness, heart failure or dehydration**: transfuse now.
- **Rapidly enlarging spleen with falling Hb** in a child with SCD: splenic sequestration, which can kill within hours.
- **SCD + fever ≥38.5 °C**: presumed pneumococcal sepsis until proven otherwise.
- **SCD + new weakness, facial droop, speech change or seizure**: stroke.
- **SCD + chest pain, cough, hypoxia, new infiltrate**: acute chest syndrome.
- **Pancytopenia, blasts, bone pain, big liver/spleen/nodes**: leukaemia. Do not give steroids before diagnosis.
- **Anaemia + bleeding + low platelets**: marrow failure, DIC, or leukaemia.

Thinking like a clinician

- **Fast breathing + pallor**: think anaemia-induced acidosis before pneumonia. Look for chest signs; if the chest is clear and the child is breathing deeply, it is metabolic.
- **Pale, jaundiced, dark urine, fever**: malaria-related haemolysis (or G6PD after a drug). If retics are high, the marrow is working.
- **Pale, not jaundiced, microcytic, eats soil, drinks lots of cow's milk**: iron deficiency. Check for hookworm.
- **Pale and macrocytic**: on AZT? folate deficiency? SCD without folate?
- **SCD with a sudden drop in Hb**: check the **spleen** and the **reticulocytes**. Big spleen and high retics means sequestration. Normal spleen and retics below 1% means aplastic crisis (parvovirus). High retics, jaundice and no splenomegaly means hyperhaemolysis or malaria.
- **"Hb 5 g/dL but the child is playing"** = chronic, compensated anaemia. TRACT tells us that immediate transfusion is not needed, but you must **recheck Hb and watch for danger signs**.
- Common student mistake: **giving a fluid bolus to a "shocked" child with severe anaemia.** Cold peripheries and tachycardia here are from anaemia and heart failure. Fluids worsen heart failure. Give **blood**.

Clinical workup

Step 0: First 5 minutes

1. **ETAT triage**: Airway, Breathing (RR, SpO_2, recession, **deep breathing**), Circulation (CRT, pulse, cold hands), Coma/Convulsion, Dehydration. Severe palmar pallor is a **priority sign**.
2. **Oxygen** if SpO_2 <90% or respiratory distress (WHO: aim SpO_2 ≥90%, ≥94% if emergency signs).
3. **IV access** and at once take blood for **Hb (HemoCue), malaria RDT, blood glucose, group and cross-match**. Draw the cross-match sample **before** any transfusion.
4. **Check glucose**. If <2.5 mmol/L (WHO; <3 mmol/L in SAM, and many units treat below 3), give **10% dextrose 5 mL/kg IV**.
5. **Shock with severe anaemia**: do not give repeated crystalloid boluses. Request **blood urgently** (O-negative or uncrossmatched group-specific in a true emergency). In shock from blood loss (e.g. trauma) crystalloid 10–20 mL/kg while blood is coming is acceptable.
6. **Convulsing**: treat (diazepam 0.5 mg/kg rectally or 0.3 mg/kg IV, or midazolam).
7. **Start treatment of the cause** in parallel: **IV artesunate** if malaria positive with severe features, **ceftriaxone** if septic or SCD with fever.

Step 1: Focused history

- Onset and speed; fever; urine colour; breathing; feeding; swelling.
- Malaria: recent episodes, treatment taken, ITN use.
- Diet history (age, breastfeeding, milk intake, iron-rich foods), pica.
- SCD: known status, previous crises and transfusions, baseline Hb, hydroxyurea, penicillin and folic acid adherence.
- Drugs and herbs; HIV status of child and mother; TB contact.
- **Immunisation card** (PCV, pentavalent, measles), **growth** (child health card), development.
- Previous transfusions and reactions.

Step 2: Focused examination

- **General**: conscious level, work of breathing, palmar pallor, jaundice, oedema, dysmorphism.
- **Vitals**: HR, RR, SpO_2, temperature, CRT, BP (normal values: HR 100–160/min infants, 80–130/min at 2–5 years; RR <50/min at 2–12 months, <40/min at 1–5 years).
- **Growth**: weight, height/length, MUAC, weight-for-height Z-score. **SAM** changes the transfusion plan.
- **Cardiovascular**: gallop, flow murmur (soft ejection systolic murmur from anaemia itself), liver span, JVP in older children.
- **Abdomen**: liver, **spleen (measure in cm below the costal margin and mark it with a pen** so the next person can see if it grows), masses.
- **Bleeding and lymph nodes; bones** (tenderness, dactylitis).
- Document key negatives: "no deep breathing, CRT <2 s, no hepatomegaly, alert" for a stable child with low Hb.

Step 3: Bedside tests

Test	What it tells you
HemoCue Hb (or Hct)	Severity in 1 minute; repeat after transfusion
Malaria RDT (± blood film)	Malaria; HRP2 RDTs stay positive for 2–4 weeks after treatment
Random blood glucose	Hypoglycaemia in severe malaria or sepsis
Sickling test / HbS point-of-care test (e.g. HemoTypeSC, Sickle SCAN where available)	Screens for SCD; a positive test needs confirmation by electrophoresis or HPLC
Urine dipstick	Haemoglobinuria (haemolysis), haematuria (schistosomiasis), protein
Pulse oximetry	Hypoxia: ACS, pneumonia, heart failure
Bedside clotting test (whole blood in a dry glass tube; no clot by 7 minutes = coagulopathy)	DIC, snakebite, severe sepsis

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Full blood count with indices	Every anaemic child	Hb, **MCV, MCH**, RDW, WBC, platelets	Classifies the anaemia; pancytopenia means marrow failure or leukaemia. Available at most HC IVs and hospitals

Test	When	Expected / abnormal finding	How it changes management
Peripheral blood film	Unexplained anaemia, abnormal FBC	Sickle cells and target cells (SCD), pencil cells (iron), blasts (leukaemia), spherocytes, fragmented cells, parasites	Points to the cause. Hospitals with a lab technologist
Reticulocyte count	Any unclear or acute anaemia, SCD crisis	**High (>2–3%)**: haemolysis or bleeding; **low (<1%)**: production failure (aplastic crisis, deficiency)	Separates sequestration from aplastic crisis. RRH/NRH, some hospitals
Group and cross-match	Before any transfusion	Compatible units	Essential. HC IV and above, depending on the blood fridge and supply
Hb electrophoresis / HPLC	Suspected SCD or thalassaemia	HbSS, HbSC, HbS-β-thal; HbF %. Do it **before transfusion** or 3 months after	Confirms SCD and enrols the child in an SCD clinic. Done at regional/national labs (CPHL, Mulago) and via DBS from **EID/newborn screening**
Serum ferritin	Microcytic anaemia	**<12 µg/L (<5 years) / <15 µg/L (≥5 years)** = iron deficiency. In infection/inflammation use <30 µg/L or <70 µg/L (WHO 2020)	Confirms iron deficiency. Mainly RRH, NRH and private labs
Stool microscopy	Iron deficiency, abdominal symptoms	Hookworm ova, *S. mansoni*	Deworm, praziquantel
Urine microscopy	Haematuria, school child in endemic area	*S. haematobium* ova	Praziquantel 40 mg/kg
HIV test	Every child with unexplained or severe anaemia	Positive	ART, cotrimoxazole, avoid AZT if anaemic
Bilirubin, LDH, DAT (Coombs)	Suspected haemolysis	Raised unconjugated bilirubin and LDH; DAT positive in autoimmune haemolysis	Steroids for autoimmune haemolysis; RRH/NRH
G6PD screen	Haemolysis after drug or fava beans	Low activity (test **after** recovery, as retics have higher G6PD)	Avoid oxidant drugs
Bone marrow aspirate	Pancytopenia, blasts, unexplained hepatosplenomegaly	Leukaemia, aplasia, leishmaniasis, infiltration	Referral to Mulago/UCI
Blood culture	SCD with fever, very sick child	Pneumococcus, non-typhoidal *Salmonella*, *Staph aureus*	Tailors antibiotics; available at RRH/NRH
Chest X-ray	SCD with chest signs, heart failure	New infiltrate (ACS), cardiomegaly	ACS treatment; heart failure
Transcranial Doppler (TCD)	Annually in HbSS aged 2–16 years	Time-averaged mean velocity **>200 cm/s abnormal**, 170–199 conditional	Start hydroxyurea or chronic transfusion to prevent stroke. Available at Mulago SCD clinic and a few other centres

Step 5: Putting it together

Model one-liner: "A 2-year-old boy, 10 kg, from Wakiso, with 3 days of fever, now has **Hb 3.8 g/dL**, a positive malaria RDT, **deep breathing (RR 56)** and a tender liver 4 cm below the costal margin. This is **severe malarial anaemia with respiratory distress and heart failure**. He needs **oxygen, IV artesunate, and packed cells 10**

mL/kg (or whole blood 20 mL/kg) over 3–4 hours with furosemide 1 mg/kg at the start."

Problem list: (1) severe anaemia with decompensation, (2) severe malaria, (3) possible coexisting iron deficiency (diet: cow's milk 1 L/day), (4) WHZ –1.5, (5) immunisation up to date.

Worked example: reading an FBC

Result	Child A	Child B	Child C
Hb (g/dL)	6.8	5.2	4.1
MCV (fL)	58	82	84
Retics	1%	12%	0.2%
Other	Platelets high 520	Spleen 6 cm, was 2 cm	Known HbSS, spleen not palpable
Interpretation	**Iron deficiency** (microcytic, reactive thrombocytosis)	**Splenic sequestration** (SCD, spleen enlarging, retics high)	**Aplastic crisis** (parvovirus B19)

A **Mentzer index** (MCV ÷ RBC count in millions) **above 13 suggests iron deficiency** and below 13 suggests thalassaemia trait. It is a screening clue only.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Severe malarial anaemia	Fever, often <3 years, acute, may have jaundice or dark urine, hepatosplenomegaly	RDT/blood film
Iron deficiency	Slow onset, pica, cow's milk excess, hookworm, well compensated	Low MCV, low ferritin, high platelets
Sickle cell disease	From 6 months, dactylitis, recurrent pain, jaundice, family history	Hb electrophoresis
HIV-related anaemia	Weight loss, recurrent infections, AZT use, oral thrush	HIV test; MCV high on AZT
Leukaemia	Bone pain, bleeding, fever, nodes, hepatosplenomegaly	FBC/film (blasts), marrow
G6PD deficiency	Boy, acute haemolysis after drug, fava, infection	G6PD assay after recovery
Visceral leishmaniasis	Karamoja/North-East, long fever, massive spleen, wasting	rK39 RDT, marrow/splenic aspirate
Hookworm / schistosomiasis	Rural, barefoot, lake swimming; haematuria	Stool/urine microscopy
Anaemia of chronic disease	TB, chronic infection, chronic kidney disease	Normal/low MCV, high ferritin, low iron
Aplastic anaemia	Pancytopenia, no organomegaly	Marrow (hypocellular)
Thalassaemia trait	Mild microcytosis out of proportion to Hb, normal ferritin	Electrophoresis (HbA2 raised in β-trait)

Investigations

Test	What you are looking for	Why
Hb / FBC with MCV	Severity and type	Directs transfusion and workup
Malaria RDT/film	Parasitaemia	Treat malaria; hyperparasitaemia is an indication to transfuse at Hb 4–6

Test	What you are looking for	Why
Reticulocytes	Marrow response	Production vs destruction/loss
Film	Cell shapes, blasts, parasites	Clinches many causes
Electrophoresis	HbS	Lifelong SCD care
Ferritin	Iron stores	Confirms iron deficiency
Stool/urine microscopy	Hookworm, schistosomes	Treatable causes
HIV test	Infection	ART; AZT-related anaemia
Group and cross-match	Compatibility	Safe transfusion

Management

A. Emergency: severe anaemia (algorithm)

1. **Triage and ABC** (ETAT). Oxygen if SpO_2 <90% or distressed. Check glucose. Take blood for Hb, RDT and cross-match before anything else.
2. **Decide on transfusion** (WHO Pocket Book 2013; UCG 2023):
   - **Hb ≤4 g/dL (Hct ≤12%)**: transfuse, whatever the clinical condition.
   - **Hb 4–6 g/dL (Hct 13–18%)** with any of: **respiratory distress/acidotic breathing, shock, impaired consciousness, heart failure, clinical dehydration, very high parasitaemia** (>10% in the Pocket Book; UCG says >20%), cerebral malaria, sepsis or meningitis: transfuse.
   - **Hb 4–6 g/dL and none of these** (uncomplicated): **do not transfuse immediately**. Treat the cause, **recheck Hb within 24 hours and watch closely**. Transfuse if Hb falls below 4 g/dL or danger signs appear (TRACT).
3. **Choose the product and volume** (table B below).
4. **Treat the cause at the same time**: artesunate for severe malaria, antibiotics for sepsis, stop oxidant drugs.
5. **Monitor** during and after transfusion (section C).
6. **Recheck Hb after transfusion** (at least 4–6 hours later, usually the next morning). If Hb is still as low and the child is still symptomatic, repeat the transfusion.
7. **Plan for discharge**: iron after the acute illness, deworming, malaria prevention, follow-up in 2 weeks and again at 1–3 months.

> **RED FLAG**
> Never give a fluid bolus to a child with severe anaemia and signs of heart failure. **In SAM, don't bolus at all unless the child is in shock**, and then use the WHO SAM shock protocol (15 mL/kg over 1 hour, and transfuse if no improvement).

B. Blood products and volumes

Product	Dose	Rate	Notes
Whole blood	**20 mL/kg**	Over **3–4 h**	UCG: available from HC IV. Hct ~35%. Avoid in severe chronic anaemia with incipient heart failure (volume overload): use packed cells
Packed red cells	**10 mL/kg** (UCG: 10–15 mL/kg)	Over **3–4 h**	Hct ~55%. Preferred in heart failure, SCD, chronic anaemia

Product	Dose	Rate	Notes
TRACT higher volume	Whole blood **30 mL/kg** (or packed cells 15 mL/kg)	Over 3–4 h	In TRACT, 30 mL/kg reduced 28-day mortality in **afebrile** children with complicated severe anaemia, but not in febrile children (possible harm). Not yet in UCG; follow your unit
SAM	**10 mL/kg** packed cells or whole blood	Over **3 h**	Only if Hb <4, or 4–6 with respiratory distress. No routine furosemide. **Do not repeat within 4 days**
Sequestration crisis (SCD)	**5–10 mL/kg** packed cells, repeat as needed	Over 2–4 h	Spleen releases trapped cells as it shrinks; aim Hb **7–8 g/dL**, not higher (hyperviscosity)
Platelets	10–15 mL/kg (or 1 unit per 10 kg)	Over 30 min	Bleeding with platelets <50 × 10^9/L, or <10 × 10^9/L without bleeding (leukaemia). Referral hospitals
Fresh frozen plasma	10–15 mL/kg	Over 30–60 min	DIC or coagulopathy with bleeding. Referral hospitals
Cryoprecipitate	5 mL/kg (or 1 unit per 5–10 kg)	Over 30 min	Fibrinogen <1–1.5 g/L with bleeding. NRH and a few RRH

How much will it raise the Hb? Roughly **8 mL/kg of whole blood or 4 mL/kg of packed cells raises Hb by 1 g/dL**. Formula: volume (mL) = weight (kg) × desired rise in Hb (g/dL) × 4 (packed cells) or × 8 (whole blood).

DRUG DOSING

Furosemide with transfusion: give **1 mg/kg IV at the start** if the child has signs of heart failure (WHO Pocket Book). Give it into the patient, never into the blood bag. If overload develops during transfusion: slow or stop, sit up, oxygen, furosemide **1–2 mg/kg IV (Pocket Book max 20 mg total)**. Routine furosemide for every transfusion is **not** recommended (UCG 2023).

UGANDA CONTEXT

The **WHO Pocket Book (2013) and UCG 2023 are the guidelines Ugandan wards follow**: 20 mL/kg whole blood or 10 mL/kg packed cells over 3–4 hours. TRACT (run in Mbale, Soroti, Mulago and Malawi) showed that **30 mL/kg** saved lives in afebrile children with severe anaemia, and that **not transfusing** uncomplicated Hb 4–6 g/dL is safe if Hb is monitored. Some Ugandan units have adopted the TRACT findings. Ask your firm which volume they use, and always write the volume in **mL**, not "1 unit".

C. Giving the transfusion safely

1. **Indication documented** with Hb and clinical signs; consent from the parent (and explain risks).
2. **Correct patient**: check name, number, group and unit number **at the bedside with two people**.
3. **Inspect the bag**: no leaks, not expired, no pink plasma, no clots, red cells not purple-black.
4. **Timing**: start within **30 minutes of removing from the fridge** (UCG) and complete within **4 hours** of starting. Do not leave blood out of the fridge for more than 2 hours before starting (Pocket Book). Do not warm it on a radiator or in water.
5. **Use a blood giving set with a filter**, a burette or infusion pump for small volumes, and a splinted arm. Do not add drugs to blood.
6. **Baseline observations** (temperature, pulse, RR, BP), then **at 15 minutes**, **every 30 minutes**, and **at the end** (UCG minimum: before, 15 min, end). Stay with the child for the first 15 minutes.
7. **Record**: start and end time, volume, unit number, any reaction.
8. **After**: reassess clinically; recheck Hb; repeat transfusion only if still indicated.

D. Transfusion reactions (UCG 2023, adapted for children)

Category	Signs	Likely cause	Management
1. Mild	Localised urticaria, itch; or mild fever <38.9 °C with no other signs	Minor allergy; febrile non-haemolytic reaction	**Pause** transfusion, check vitals. Antihistamine (**chlorphenamine** 0.1 mg/kg IV/oral, or cetirizine/loratadine; UCG lists promethazine IM, 5 mg at 1–5 years, 6.25–12.5 mg at 5–10 years; avoid promethazine under 2 years). For fever give **paracetamol 15 mg/kg**. If stable, **restart slowly** and observe. If no improvement in 30 min or worsening: treat as category 2
2. Severe / life-threatening	Fever ≥39 °C, rigors, hypotension (fall >20% in SBP), tachycardia, dyspnoea, wheeze/stridor, chest/loin/back pain, headache, restlessness, **haemoglobinuria**, unexplained bleeding (DIC), widespread rash, airway oedema	**Acute haemolytic** (ABO mismatch), **bacterial contamination**, **anaphylaxis**, **TACO** (circulatory overload), **TRALI** (lung injury)	**STOP and do not restart.** Keep the IV line with **normal saline**. ABC, high-flow oxygen. **Anaphylaxis**: **adrenaline 0.01 mg/kg IM** (0.01 mL/kg of 1:1000, max 0.5 mg), hydrocortisone 4 mg/kg IV, antihistamine, salbutamol neb 2.5–5 mg. **Hypotension**: NS 20 mL/kg (UCG 20–30 mL/kg) unless the cause is overload. **TACO**: sit up, furosemide 1 mg/kg IV, no fluids. Suspected contamination: blood cultures and broad-spectrum antibiotics (ceftriaxone). Check urine for haemoglobin, start a fluid balance chart. **Send the bag, giving set, fresh urine and new blood samples (clotted and EDTA, from the other arm) to the blood bank** and report

CLINICAL PEARL

A **conscious** child with an ABO-incompatible transfusion may react after only **5–10 mL**. In an **unconscious** child, hypotension, hypoxia or unexplained bleeding may be the only signs. Most fatal haemolytic reactions are due to **clerical errors** at sample collection or at the bedside.

Delayed complications: transfusion-transmitted infection (HIV, hepatitis B/C, syphilis, malaria), delayed haemolytic reaction (5–10 days later, especially in multiply transfused SCD children), **iron overload** with repeated transfusions (ferritin >1000 µg/L; chelation with deferasirox at specialist centres).

E. Iron deficiency anaemia

Drug	Dose	Route	Frequency	Duration	Notes
Ferrous sulphate	**UCG 2023**: 5 mg/kg of the salt (max 200 mg/dose) ≈ 3–6 mg/kg/day **elemental iron** in total	Oral	**8-hourly** (or the daily dose once a day, which is better tolerated)	Until Hb normal, **then 3 more months** to refill stores (UCG: 6 months to 1 year if stores very low)	200 mg tablet = 60–65 mg elemental iron. Give between meals, not with milk or tea; vitamin C helps. Black stools are expected; constipation or epigastric pain are common
Iron syrup (e.g. ferrous fumarate/sulphate drops)	3 mg/kg/day elemental iron (max 200 mg/day in older children; 60 mg in infants)	Oral	Once or twice daily	As above	For infants; overdose is dangerous, keep it away from children
Albendazole	200 mg (12–23 months); **400 mg** (≥2 years)	Oral	Single dose	Every 6 months	Hookworm. Mebendazole 500 mg single dose is an alternative (≥1 year)

Drug	Dose	Route	Frequency	Duration	Notes
Praziquantel	**40 mg/kg**	Oral	Single dose	—	Schistosomiasis (≥4 years in mass programmes; clinical use younger with paediatric formulation)
Folic acid	2.5–5 mg	Oral	Daily	2–4 months	If folate deficiency or mixed deficiency

- **Expected response**: reticulocytes rise in **5–10 days**; Hb rises about **1–2 g/dL in 2–4 weeks** (UCG: rises in 2–3 weeks, normal by 2 months). **No rise after 4 weeks** means poor adherence, ongoing loss, malabsorption, wrong diagnosis (thalassaemia trait, chronic disease) or an infection.
- **Diet advice**: exclusive breastfeeding to 6 months; iron-rich complementary foods (liver, meat, fish, small fish *mukene*, beans, dark green leaves), **limit cow's milk to about 500 mL/day** after 12 months, no tea with meals.
- **Iron and malaria**: in malaria-endemic areas, give iron **together with malaria prevention** (ITN) and **treatment**. Postpone iron during acute severe illness and in the **stabilisation phase of SAM** (start in rehabilitation).
- **Parenteral iron** is rarely needed in children and is a specialist decision (anaphylaxis risk).

F. Sickle cell disease: long-term care (SCD clinic)

Intervention	Dose (child)	Route / frequency	Duration	Source
Penicillin V	**3 months–3 years: 125 mg**; **3–5 years: 250 mg**	Oral, **12-hourly**	Until **5 years** (continue after 5 if previous invasive pneumococcal disease or splenectomy)	UCG 2023. Alternative where adherence is poor: benzathine penicillin IM monthly (specialist choice). Erythromycin 125–250 mg 12-hourly if allergic
Folic acid	**5 mg** (UCG: for life; some clinics give 1–2.5 mg in infants)	Oral, daily	Life	UCG 2023
Malaria chemoprevention	**SP**: ½ tab at 2–5 y, 1 tab at 5–10 y, 2 tabs at 10–15 y, 3 tabs >15 y (1 tab = 500/25 mg)	Oral, **monthly**	Life (with ITN)	UCG 2023. Avoid SP if sulphonamide allergy or on cotrimoxazole prophylaxis. Some Ugandan SCD clinics use monthly dihydroartemisinin-piperaquine instead (VERIFY local practice)
Vaccines	Full **UNEPI** schedule (pentavalent incl. Hib, **PCV**, measles-rubella, etc.) plus **PPSV23 at 2 years** (repeat after 3–5 years), meningococcal and influenza if available, hepatitis B check	IM	—	UCG 2023
Hydroxyurea	Start **20 mg/kg once daily** (UCG; REACH 15–20 mg/kg); escalate by **5 mg/kg every 8 weeks** to the **maximum tolerated dose, max 35 mg/kg/day**	Oral, once daily (capsules can be opened into water)	Long term; benefit is lost if stopped	UCG 2023; NHLBI 2014; REACH; NOHARM

Intervention	Dose (child)	Route / frequency	Duration	Source
Transcranial Doppler	—	Annually, 2–16 years	—	Abnormal (>200 cm/s): hydroxyurea at MTD or chronic transfusion

Hydroxyurea in detail

- **Who** (UCG 2023): children **≥9 months** with **>3 crises a year**, pain interfering with daily life, **abnormal TCD (>200 cm/s)**, recurrent or severe **acute chest syndrome**, or **stroke**. International guidance (NHLBI 2014, ASH) and the African trials support **offering it to every child with HbSS or HbS-β⁰ from 9 months**, whatever the severity. Many Ugandan SCD clinics now follow the broader approach where supply allows.
- **Before starting**: FBC, retics, renal and liver function, HbF if available; counsel the family. A senior clinician makes the decision (UCG).
- **Monitoring**: FBC every **2–4 weeks during escalation**, then every **2–3 months**.
- **Hold if** (toxicity): **ANC <1.0 × 10^9/L**, **platelets <80 × 10^9/L**, Hb falling below 5 g/dL or retics <80 × 10^9/L when Hb is low. Restart at a lower dose (by 2.5–5 mg/kg) once counts recover (usually 1–2 weeks).
- **Expected effects**: MCV rises (good adherence marker), HbF rises, fewer pain crises, ACS, transfusions and deaths. **NOHARM (Uganda, 2017)** showed **no increase in malaria**; **REACH** (Uganda, Kenya, DRC, Angola) showed fewer events and deaths, and **dose escalation to MTD** gave better results than a fixed dose.
- **Side effects**: myelosuppression (dose-related, reversible), nail and skin darkening, GI upset. Advise adolescents on **contraception** (teratogenic in animals; stop 3 months before conception, as in the obs topic).

G. Sickle cell disease: acute complications

Complication	Recognise	Manage
Fever ≥38.5 °C	Any fever, especially <5 years	Blood culture, RDT, FBC, CXR/urine as indicated; **ceftriaxone 80–100 mg/kg IV once daily (max 4 g)** within 1 hour; add artesunate if malaria. Admit if unwell, <1 year, Hb drop, or temperature ≥39.5 °C
Vaso-occlusive (painful) crisis	Bone, back, chest or abdominal pain; dactylitis (hand-foot syndrome) in <2 years	**Analgesia within 30 minutes** and reassess every 30–60 min: **paracetamol 15 mg/kg 6-hourly** (max 60 mg/kg/day; UCG 10–15 mg/kg) + **ibuprofen 5–10 mg/kg 8-hourly** (max 30 mg/kg/day; not if dehydrated or kidney disease) → **oral morphine 0.2–0.4 mg/kg 4-hourly** or **IV morphine 0.1 mg/kg** titrated (UCG 0.1–0.2 mg/kg). Avoid pethidine and codeine under 12 years. Hydration at maintenance (oral or IV; do not overload). Warm compresses (not cold). Oxygen only if SpO_2 <95%. Laxative with morphine (bisacodyl 2.5–5 mg)
Acute chest syndrome	New infiltrate on CXR + fever, cough, chest pain, tachypnoea or hypoxia	Oxygen to SpO_2 ≥95%, analgesia (avoid oversedation), **incentive spirometry** (or blowing balloons) every 2 h awake, **ceftriaxone 80–100 mg/kg** + a macrolide (**azithromycin 10 mg/kg once daily** or erythromycin 10–12.5 mg/kg 6-hourly; UCG lists 5–10 mg/kg) for 7–10 days, salbutamol if wheeze, **restricted IV fluids** (do not overload). **Transfuse** if Hb falls or <9 g/dL (UCG target ~10–11 g/dL, never above 11); **exchange transfusion** if severe or worsening
Splenic sequestration	Pallor, weakness, **rapidly enlarging spleen**, Hb drop ≥2 g/dL, high retics; can cause shock	**Urgent transfusion** of packed cells **5–10 mL/kg aliquots**; aim Hb 7–8 g/dL; avoid hyperviscosity. Mark the spleen; teach parents to palpate it. After 2 episodes or one life-threatening one, consider **splenectomy** when well (UCG: avoid splenectomy in the acute phase)

Complication	Recognise	Manage
Aplastic crisis	Pallor, tiredness after a febrile illness, **retics <1%**, spleen not bigger	Transfuse if symptomatic or Hb very low (packed cells 10 mL/kg). Usually recovers in 7–10 days. Isolate from pregnant women (parvovirus B19)
Stroke	Hemiparesis, speech or vision change, seizure, reduced consciousness	ABC, **oxygen** (SpO_2 >94%), check glucose, **urgent transfusion**: **exchange transfusion** is preferred; if not available, simple transfusion to Hb **≤10–11 g/dL** (UCG: transfuse if Hb <9 g/dL). Refer for neuroimaging. Then **chronic transfusion or hydroxyurea at MTD** for secondary prevention
Priapism	Painful erection >2–4 h	Hydration, analgesia, urinate; urology referral for aspiration if >4 h
Mesenteric crisis / acute abdomen	Distension, absent bowel sounds, pain	Nil by mouth, NG tube, IV fluids, ceftriaxone + metronidazole 7.5–10 mg/kg 8-hourly, transfusion, surgical review
Osteomyelitis / septic arthritis	Fever, persistent focal bone pain (vs crisis), swelling	*Salmonella* and *Staph*: **ceftriaxone 50–80 mg/kg** or cloxacillin 50 mg/kg 6-hourly (max 2 g/dose), for at least 21 days (UCG)

Transfusion indications in SCD (UCG 2023): acute fall in Hb ≥2 g/dL below baseline (or Hb <5 g/dL), sequestration (splenic or hepatic), aplastic crisis, hyperhaemolysis, stroke, ACS, multi-organ failure, before surgery or general anaesthesia. **Do not transfuse** a well child at steady state (Hb 6–8 g/dL) or for an uncomplicated painful crisis. Use **packed cells**, not whole blood, where possible, and **never raise Hb above about 11 g/dL** (hyperviscosity, stroke).

H. Follow-up after severe anaemia

- Review at **2 weeks** and **1–3 months**: Hb, adherence to iron, growth.
- **Malaria prevention**: ITN; post-discharge malaria chemoprevention is recommended by WHO for children admitted with severe anaemia in high-transmission areas (monthly antimalarial courses for 3 months), and Uganda has piloted it (VERIFY local rollout).
- Screen siblings for SCD; refer to an SCD clinic; counsel on genetics.

Complications

Of anaemia	Of SCD	Of transfusion
High-output heart failure	Stroke (up to 10% of HbSS children without screening)	Acute haemolytic reaction (ABO)
Lactic acidosis, death	Acute chest syndrome	Febrile, allergic, anaphylaxis
Impaired neurodevelopment (iron deficiency)	Splenic sequestration, aplastic crisis	TACO, TRALI
Poor school performance	Invasive pneumococcal/Salmonella infection	Bacterial contamination, sepsis

Of anaemia	Of SCD	Of transfusion
Readmission and post-discharge death	Avascular necrosis of the femoral head, leg ulcers, priapism	HIV, hepatitis B/C, syphilis, malaria transmission
	Gallstones, nephropathy, growth delay, delayed puberty	Iron overload, alloimmunisation

Ward-round & exam pearls

Model case presentation

"**Amina is a 3-year-old girl weighing 11.5 kg** (weight-for-age Z −1.6, height-for-age Z −1.0, WHZ −1.4, MUAC 13.6 cm), a known **HbSS** patient attending Mulago SCD clinic. She is fully **immunised for age per UNEPI** including PCV, and on **penicillin V 125 mg twice daily** and folic acid, but she missed her last 2 clinic visits. She eats family foods 3 times daily and still breastfeeds. **HIV status: negative** (mother negative in ANC). She presented with **2 days of fever and progressive weakness**. On examination she is **very pale**, RR 48 with no recession, HR 160, CRT 3 seconds, and her **spleen is 7 cm below the costal margin, up from 2 cm at her last visit**. **Hb is 3.9 g/dL** (baseline 7.5) and retics are 14%. The RDT is negative. My working diagnosis is **acute splenic sequestration crisis**, with possible sepsis. I have taken a blood culture, given ceftriaxone 100 mg/kg, and requested packed cells, which I plan to give as **5–10 mL/kg aliquots aiming for Hb 7–8 g/dL**, with spleen size marked and checked hourly."

Viva questions (with answers)

1. **What are the indications for transfusion in a child?** Hb ≤4 g/dL; Hb 4–6 g/dL with respiratory distress, shock, impaired consciousness, heart failure, dehydration or high parasitaemia; acute blood loss >20–30% of volume; SCD-specific indications.
2. **How much blood do you give?** Whole blood 20 mL/kg or packed cells 10 mL/kg over 3–4 hours; 10 mL/kg over 3 h in SAM.
3. **How do you tell sequestration from aplastic crisis?** Spleen size and reticulocytes: sequestration = enlarging spleen, high retics; aplastic = no splenic change, retics <1%.
4. **Why penicillin in SCD?** Functional asplenia from infancy; pneumococcal sepsis kills; penicillin prophylaxis (with PCV) cuts invasive pneumococcal disease by about 80%.
5. **How does hydroxyurea work and how do you monitor it?** Raises HbF and lowers WBC; FBC every 2–4 weeks during escalation; hold if ANC <1.0 or platelets <80.
6. **A child develops fever and back pain 10 minutes into a transfusion. What do you do?** Stop the transfusion, keep the line open with saline, ABC and vitals, check identity against the bag, treat shock, send the bag and samples to the blood bank, check urine for haemoglobin.

OSCE tips

- In a "pale child" station, say **"palmar pallor"**, examine for **spleen and liver**, and **look for signs of heart failure**: these decide how to transfuse.
- In a counselling station on SCD, cover: genetics (autosomal recessive, 1 in 4 risk for each pregnancy of two carriers), fever = hospital, penicillin, folic acid, malaria prevention, fluids, how to feel the spleen, hydroxyurea and clinic follow-up.

Mnemonics & memory aids

- **Transfuse in Hb 4–6 if "RISC-DH"**: **R**espiratory distress, **I**mpaired consciousness, **S**hock, **C**ardiac failure, **D**ehydration, **H**yperparasitaemia.
- **Blood volume "20-10-10"**: **20** mL/kg whole blood, **10** mL/kg packed cells, **10** mL/kg in SAM.
- **"8 and 4"**: 8 mL/kg whole blood or 4 mL/kg packed cells raises Hb by 1 g/dL.
- **SCD clinic package "PFMVH"**: **P**enicillin, **F**olic acid, **M**alaria prevention, **V**accines, **H**ydroxyurea (and TCD).
- **Sequestration vs aplastic: "Big spleen, busy marrow" vs "Small retics, sleeping marrow".**

High-yield summary

Numbers & doses to know

Item	Value
Anaemia cut-off 6–23 mo / 24–59 mo / 5–11 y (WHO 2024)	<10.5 / <11.0 / <11.5 g/dL
Severe anaemia (<5 y / ≥5 y)	<7 / <8 g/dL
Transfuse regardless of signs	Hb ≤4 g/dL (Hct ≤12%)
Transfuse with danger signs	Hb 4–6 g/dL
Whole blood / packed cells	20 mL/kg / 10 mL/kg over 3–4 h
SAM	10 mL/kg over 3 h; no repeat within 4 days
Start / finish blood	Within 30 min of leaving fridge / within 4 h
Furosemide with transfusion in heart failure	1 mg/kg IV at start
Adrenaline in anaphylaxis	0.01 mg/kg IM (max 0.5 mg)
Ferrous sulphate (UCG)	5 mg/kg salt 8-hourly (max 200 mg/dose); 3 months after Hb normal
Albendazole	200 mg (1–2 y), 400 mg (≥2 y)
Penicillin V in SCD	125 mg bd (3 mo–3 y); 250 mg bd (3–5 y)
Folic acid in SCD	5 mg daily
Hydroxyurea	Start 20 mg/kg/day from 9 months; +5 mg/kg every 8 weeks; max 35 mg/kg/day
Hydroxyurea hold	ANC <1.0, platelets <80 × 10^9/L
Ceftriaxone in SCD fever	80–100 mg/kg IV once daily (max 4 g)
Abnormal TCD	>200 cm/s
Target Hb after SCD transfusion	Do not exceed ~11 g/dL

Classic exam traps

- **Trap**: "Hb 5 g/dL, so transfuse." **Right**: in an alert child with no distress, TRACT and WHO say monitor, treat the cause, and transfuse only if Hb falls below 4 or danger signs appear.
- **Trap**: Fluid bolus for "shock" in severe anaemia. **Right**: give blood; fluids worsen heart failure.
- **Trap**: Giving 20 mL/kg whole blood quickly to a child with SAM. **Right**: 10 mL/kg over 3 h.
- **Trap**: Transfusing an SCD child up to Hb 13 g/dL "to be safe". **Right**: never above ~11 g/dL (hyperviscosity, stroke).

- **Trap**: Starting iron in a child with severe malaria on day 1. **Right**: treat malaria first; start iron when the acute illness has settled, alongside malaria prevention.
- **Trap**: Reassuring because "the RDT is positive, so the anaemia is from malaria". **Right**: most anaemic Ugandan children have more than one cause; check MCV, diet, worms, HIV, SCD.
- **Trap**: Waiting to see whether a febrile SCD child has malaria before giving antibiotics. **Right**: ceftriaxone within 1 hour after a blood culture.

Questions seniors ask

1. **Why is this child breathing fast?** Usually metabolic acidosis from anaemia (deep breathing, clear chest), sometimes heart failure or pneumonia. The treatment differs.
2. **Why packed cells rather than whole blood?** Less volume for the same Hb rise, so less risk of overload in heart failure, chronic anaemia and SCD.
3. **What is the most common cause of microcytic anaemia here, and how do you confirm it?** Iron deficiency; low MCV/MCH, high RDW, low ferritin (<12 µg/L under 5 years) and a response to iron.
4. **What did TRACT show?** No benefit of immediate transfusion in uncomplicated Hb 4–6 g/dL; 30 mL/kg reduced mortality in afebrile complicated severe anaemia.
5. **When do you start hydroxyurea and what dose?** From 9 months; 20 mg/kg/day (UCG), escalating to MTD up to 35 mg/kg/day.
6. **What did NOHARM show?** Hydroxyurea at 20 mg/kg/day did not increase malaria in Ugandan children with SCA.
7. **What do you check before you hang the blood?** Patient ID against the bag and form (two people), expiry, bag integrity and appearance, time out of the fridge, baseline vitals, signs of heart failure.

SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	WHO 2024 Hb cut-offs for anaemia in children (6–23 mo, 24–59 mo, 5–11 y, 12–14 y)?	**<10.5, <11.0, <11.5 and <12.0 g/dL** respectively. Severe anaemia: **<7 g/dL** under 5 years, **<8 g/dL** from 5 years. (UCG 2023 still uses <11 g/dL for 6 mo–5 y.)
2	Absolute indication for transfusion in a child (WHO Pocket Book / UCG 2023)?	**Hb ≤4 g/dL (Hct ≤12%)**, whatever the clinical condition.
3	When do you transfuse a child with Hb 4–6 g/dL?	If any of: **respiratory distress/acidotic breathing, impaired consciousness, shock, heart failure, clinical dehydration, very high parasitaemia** (>10% Pocket Book; UCG >20%), cerebral malaria, sepsis or meningitis. Mnemonic **RISC-DH**.
4	Standard transfusion volumes in children?	**Whole blood 20 mL/kg** or **packed cells 10 mL/kg** (UCG 10–15 mL/kg) over **3–4 hours**.
5	Transfusion in severe acute malnutrition?	Only if Hb <4 g/dL or 4–6 g/dL with respiratory distress: **10 mL/kg packed cells or whole blood over 3 h**, no routine furosemide, **do not repeat within 4 days**.
6	Rule of thumb: how much blood raises Hb by 1 g/dL?	About **8 mL/kg whole blood** or **4 mL/kg packed cells**.
7	What did the TRACT trial (Uganda/Malawi, NEJM 2019) show?	1. In **uncomplicated** severe anaemia (Hb 4–6, no severity signs), **immediate transfusion did not improve survival** vs monitoring with a trigger (Hb <4 or new signs). 2) In complicated severe anaemia, **30 vs 20 mL/kg** reduced 28-day mortality in **afebrile** children but not in febrile children.
8	Time limits for giving a unit of blood (UCG 2023)?	Start within **30 minutes** of removing from the fridge; complete within **4 hours** of starting. Store at 2–6 °C. Do not warm improvised.
9	When are vital signs recorded during a transfusion?	**Before**, at **15 minutes**, every **30 minutes** (Pocket Book), at the **end**, and whenever a reaction is suspected. Observe the child closely for the first 15 minutes.
10	First 4 steps when a severe transfusion reaction is suspected?	1. **Stop** the transfusion (do not restart) 2) keep IV access with **normal saline** 3) **check patient ID vs bag** 4) ABC/resuscitate; then send bag, giving set, fresh urine and new blood samples to the blood bank and report.
11	Dose of adrenaline for anaphylaxis during transfusion in a child?	**0.01 mg/kg IM** (0.01 mL/kg of 1:1000), **max 0.5 mg**; repeat after 5 min if needed. Plus oxygen, fluids, hydrocortisone 4 mg/kg IV, antihistamine.
12	Furosemide and transfusion: when and how much?	Only if signs of **heart failure**: **1 mg/kg IV at the start** (into the child, never into the bag). For overload during transfusion, 1–2 mg/kg (Pocket Book max 20 mg). Not routine for every transfusion (UCG).
13	Classify anaemia by MCV with Ugandan examples.	**Microcytic**: iron deficiency, thalassaemia trait. **Normocytic**: malaria, SCD, blood loss, HIV, marrow infiltration. **Macrocytic**: folate/B12 deficiency, **zidovudine**, hydroxyurea, reticulocytosis.
14	Ferritin cut-off for iron deficiency in children (WHO 2020)?	**<12 µg/L under 5 years**, **<15 µg/L from 5 years** in healthy children; with infection/inflammation **<30 µg/L** (under 5) or **<70 µg/L** (5 years and older).

#	Question	Answer
15	Oral iron regimen for iron deficiency (UCG 2023)?	**Ferrous sulphate 5 mg/kg (of salt) 8-hourly, max 200 mg/dose** (≈3–6 mg/kg/day elemental iron), until Hb normal then **3 more months** (UCG: 6–12 months for stores). Plus **albendazole** (200 mg at 1–2 y, 400 mg ≥2 y).
16	Expected response to oral iron?	Reticulocytes rise in **5–10 days**; Hb rises **1–2 g/dL in 2–4 weeks**. No response after 4 weeks: check adherence, ongoing loss, wrong diagnosis (thalassaemia trait, chronic disease), infection.
17	SCD penicillin prophylaxis doses (UCG 2023)?	**Penicillin V 125 mg 12-hourly from 3 months to 3 years; 250 mg 12-hourly from 3 to 5 years.** Erythromycin if allergic.
18	Hydroxyurea in SCD: start age, dose, escalation and max?	From **9 months**; start **20 mg/kg once daily** (REACH 15–20); escalate **5 mg/kg every 8 weeks** to **maximum tolerated dose, max 35 mg/kg/day**.
19	When do you hold hydroxyurea?	**ANC <1.0 × 10^9/L**, **platelets <80 × 10^9/L**, or a significant fall in Hb with low reticulocytes. Restart at a lower dose when counts recover.
20	What did NOHARM show?	In malaria-endemic **Uganda**, hydroxyurea **20 mg/kg/day** for 12 months did **not increase malaria incidence** vs placebo, and was safe (Blood 2017).
21	Splenic sequestration vs aplastic crisis?	**Sequestration**: rapidly enlarging spleen, Hb drop ≥2 g/dL, **high retics**, may be shocked. **Aplastic** (parvovirus B19): no splenic enlargement, **retics <1%**. Both need urgent transfusion.
22	How do you transfuse in splenic sequestration and why?	Packed cells in **5–10 mL/kg aliquots**, aim Hb **7–8 g/dL**: as the spleen shrinks it **releases trapped red cells** (auto-transfusion), so a big transfusion can cause hyperviscosity.
23	Abnormal transcranial Doppler velocity in SCD and action?	Time-averaged mean velocity **>200 cm/s** (170–199 conditional). Start **hydroxyurea at MTD or chronic transfusion** to prevent stroke. Screen annually from 2 to 16 years.
24	Define acute chest syndrome and list the treatment bundle.	New pulmonary infiltrate + fever, cough, chest pain, tachypnoea or hypoxia. **Oxygen, analgesia, incentive spirometry, ceftriaxone + macrolide, cautious fluids, transfusion** (simple if Hb <9, exchange if severe).
25	Maximum target Hb after transfusion in SCD?	About **10–11 g/dL** (UCG target 11 g/dL, not higher). Higher Hb increases viscosity and stroke risk.
26	Why does SCD present after about 6 months of age?	**HbF ($\alpha_2\gamma_2$)** does not sickle and protects until the γ→β globin switch completes. Hydroxyurea works mainly by **raising HbF**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy (11 kg) from Lira is brought with 3 days of fever. He is alert and playing. RR 32/min, HR 130/min, CRT <2 s, no hepatomegaly, SpO_2 97%. Malaria RDT is positive and HemoCue Hb is 5.1 g/dL. There are no other danger signs. What is the MOST appropriate plan for his anaemia?

A Transfuse whole blood 20 mL/kg over 3–4 hours now

B Transfuse whole blood 30 mL/kg because the TRACT trial showed benefit

C Treat the malaria, do not transfuse now, recheck Hb and watch for danger signs, transfusing if Hb falls below 4 g/dL or signs appear

D Give a 20 mL/kg normal saline bolus and recheck Hb

E Start oral iron today and discharge with review in 2 weeks

2. A 3-year-old girl (12 kg) with severe malaria has Hb 4.6 g/dL, deep laboured breathing with a clear chest, HR 168/min, a gallop rhythm and a liver 5 cm below the costal margin. Packed cells are available. What is the BEST transfusion prescription?

A Packed cells 120 mL (10 mL/kg) over 3–4 h with furosemide 12 mg (1 mg/kg) IV at the start

B Whole blood 360 mL (30 mL/kg) over 1 hour

C Normal saline 240 mL (20 mL/kg) first, then whole blood

D No transfusion until Hb is below 4 g/dL

E Packed cells 240 mL (20 mL/kg) over 1 hour with furosemide 2 mg/kg into the bag

3. A 4-year-old boy with HbSS (14 kg) presents with pallor and weakness after a viral illness. HR 140, RR 30, alert. Spleen not palpable (unchanged from clinic). Hb 3.6 g/dL (baseline 7.8), reticulocytes 0.3%. What is the MOST likely diagnosis?

A Splenic sequestration crisis

B Hyperhaemolytic crisis

C Iron deficiency

D Aplastic crisis due to parvovirus B19

E Severe malaria

4. A 20-month-old with HbSS (9 kg) is brought pale and lethargic. HR 175, CRT 4 s, spleen 8 cm below the costal margin (2 cm last month). Hb 3.2 g/dL, reticulocytes 15%. Blood is being cross-matched. What is the MOST appropriate transfusion target and approach?

A Transfuse whole blood 30 mL/kg to bring Hb to 12 g/dL

B Transfuse packed cells in 5–10 mL/kg aliquots aiming for Hb about 7–8 g/dL, reassessing the spleen

C Do not transfuse; perform emergency splenectomy

D Give normal saline 20 mL/kg boluses until CRT is normal, then transfuse if still pale

E Give hydroxyurea 20 mg/kg and observe

5. Twenty minutes into a whole-blood transfusion, a 7-year-old boy (22 kg) becomes restless, complains of back pain, has a temperature of 39.4 °C and BP falls from 100/60 to 75/40 mmHg. What is the FIRST action?

A Give paracetamol 15 mg/kg and slow the transfusion

B Give chlorphenamine and continue at the same rate

C Give furosemide 1 mg/kg IV and continue the transfusion

D Send a sample for a repeat cross-match and continue while awaiting the result

E Stop the transfusion, keep the IV line open with normal saline and check the patient's identity against the bag

6. A 15-month-old girl (8.2 kg) has pallor for 2 months. She drinks about 1.2 L of cow's milk daily and eats little else; she eats soil. Hb 7.2 g/dL, MCV 56 fL, platelets 560 × 10^9/L, RDT negative. She is alert with normal vital signs. What is the MOST appropriate management?

A Transfuse packed cells 10 mL/kg

B Folic acid 5 mg daily only

C Oral iron (about 3–6 mg/kg/day elemental), albendazole 200 mg, reduce cow's milk to about 500 mL/day and add iron-rich foods, review in 2–4 weeks

D IV iron sucrose at the HC IV

E Hb electrophoresis and no treatment until the result returns

7. A 2-year-old with HbSS (10.5 kg) has a temperature of 39.2 °C for 6 hours. He is alert, HR 150, RR 34, Hb 7.0 g/dL (baseline 7.4). The RDT is negative. What should you do NEXT?

A Give paracetamol and review in the morning

B Take a blood culture and give ceftriaxone 80–100 mg/kg IV within 1 hour

C Give oral amoxicillin and discharge

D Start artemether-lumefantrine because RDTs can miss malaria

E Transfuse because Hb is low

8. A 6-year-old with HbSS (18 kg) suddenly develops right-sided weakness and slurred speech. SpO_2 93%, Hb 7.8 g/dL. Exchange transfusion is not available at your regional referral hospital today. What is the MOST appropriate immediate management? Advanced

A Give aspirin 75 mg and arrange CT next week

B Start hydroxyurea 20 mg/kg and discharge

C Give oxygen, check glucose, and give a simple transfusion of packed cells aiming for Hb about 10–11 g/dL, then refer urgently for imaging and exchange transfusion

D Transfuse whole blood 30 mL/kg aiming for Hb 14 g/dL

E Give IV mannitol and fluid restrict

9. A 4-year-old (13 kg) with severe acute malnutrition (WHZ −3.4, no oedema) on F-75 has Hb 3.6 g/dL with fast breathing. Which transfusion order is MOST appropriate?

A Packed cells or whole blood 10 mL/kg (130 mL) over 3 hours, no routine furosemide, and do not repeat within 4 days

B Whole blood 20 mL/kg over 3–4 hours with furosemide

C Whole blood 30 mL/kg over 4 hours

D Do not transfuse children with SAM

E Packed cells 10 mL/kg over 30 minutes

10. A 2-year-old with HbSS (11 kg) attends the SCD clinic. She has had 1 painful crisis this year, no stroke or acute chest syndrome, and TCD is normal. ANC 4.2, platelets 310 × 10^9/L, renal and liver function normal. The mother asks whether she should start hydroxyurea. What is the BEST response? Advanced

A No; hydroxyurea is only for children with a previous stroke

B No; hydroxyurea increases malaria risk in Uganda

C No; wait until she is 5 years old

D Yes; hydroxyurea can be offered from 9 months, starting at about 20 mg/kg once daily with FBC monitoring and dose escalation; UCG lists frequent crises as an indication, but international guidance offers it to all children with HbSS

E Yes; start at 35 mg/kg/day to get the maximum effect immediately

11. A 5-year-old on hydroxyurea 25 mg/kg/day has routine bloods: Hb 8.9 g/dL, MCV 102 fL, ANC 0.8 × 10^9/L, platelets 150 × 10^9/L. She is well. What is the MOST appropriate action? Advanced

A Increase to 30 mg/kg because MCV shows good adherence

B Stop hydroxyurea permanently

C Hold hydroxyurea, recheck FBC in about 1 week, and restart at a lower dose (e.g. 20 mg/kg) when ANC recovers above 1.0 × 10^9/L

D Start folic acid and ignore the neutrophil count

E Give G-CSF and continue the same dose

12. A 9-year-old with HbSS (24 kg) has had 2 days of fever, cough and chest pain. SpO_2 89% in air, RR 38, crackles at the right base; CXR shows new right lower lobe consolidation. Hb 6.9 g/dL (baseline 8.1). What is the MOST appropriate management?

A Oral amoxicillin and discharge with analgesia

B Oxygen, analgesia, incentive spirometry, IV ceftriaxone plus a macrolide, cautious fluids, and transfusion of packed cells

C IV fluids at twice maintenance to reverse sickling

D Morphine infusion only, avoiding antibiotics as this is a crisis not an infection

E High-dose dexamethasone

13. A 3-year-old boy (13 kg) on zidovudine-containing ART for HIV has Hb 7.4 g/dL, MCV 108 fL. Malaria RDT negative. Which is the MOST likely cause of the anaemia?

A Iron deficiency

B β-thalassaemia trait

C Hookworm infestation

D Splenic sequestration

E Zidovudine (AZT) marrow toxicity

14. Which statement about blood transfusion practice in Uganda (UCG 2023) is CORRECT?

A Blood should be started within 30 minutes of removal from the fridge and completed within 4 hours

B Blood should be warmed in a basin of hot water before use

C Furosemide should be given routinely before every transfusion

D Whole blood is only available at national referral hospitals

E Vital signs need only be taken at the end of the transfusion

15. A 10-month-old (7.8 kg) was admitted with severe malarial anaemia, transfused and treated with artesunate, and is now well with Hb 8.1 g/dL. Weight-for-length Z −1.2. What discharge plan BEST reduces his risk of readmission and death? Advanced

A Monthly blood transfusion for 3 months

B No follow-up is needed since he has recovered

C Start hydroxyurea

D Insecticide-treated net, post-discharge malaria chemoprevention where available, oral iron once recovered, nutrition counselling, and review at 2 weeks and 1–3 months

E Weekly chloroquine for life

Answers & explanations

1. C · Treat the malaria, do not transfuse now, recheck Hb and watch for danger signs, transfusing if Hb falls below 4 g/dL or signs appear

C is correct: this is **uncomplicated** severe anaemia (Hb 4–6 g/dL with no respiratory distress, shock, impaired consciousness or heart failure). WHO/UCG do not require transfusion, and TRACT showed no benefit from immediate transfusion when Hb was monitored with a trigger. A is wrong: transfusion at Hb 4–6 needs a severity feature. B is wrong: the 30 mL/kg benefit in TRACT was in **complicated** severe anaemia in afebrile children, and this child is febrile and uncomplicated. D is wrong: fluid boluses do not treat anaemia and risk heart failure. E is wrong: he needs malaria treatment and close Hb monitoring first; iron is started after the acute illness.

2. A · Packed cells 120 mL (10 mL/kg) over 3–4 h with furosemide 12 mg (1 mg/kg) IV at the start

A is correct: she has Hb 4–6 g/dL with acidotic breathing and **heart failure**, so she needs blood; packed cells 10 mL/kg over 3–4 h with **furosemide 1 mg/kg IV at the start** is the WHO Pocket Book approach. B is wrong: a large volume over 1 hour risks fatal overload in heart failure. C is wrong: a saline bolus worsens heart failure. D is wrong: Hb 4–6 with danger signs is an indication. E is wrong: 20 mL/kg of packed cells is double the dose, 1 hour is too fast, and drugs must never be added to the blood bag.

3. D · Aplastic crisis due to parvovirus B19

D is correct: a sharp fall in Hb with **reticulocytes below 1%** and no splenic enlargement after a viral illness is the classic **aplastic crisis** from parvovirus B19 arresting red cell production. A is wrong: sequestration gives a rapidly enlarging spleen and high retics. B is wrong: hyperhaemolysis gives high retics and rising jaundice. C is wrong: iron deficiency develops slowly and is unusual in SCD; it does not drop Hb by 4 g/dL in days. E is wrong: malaria gives haemolysis with a raised reticulocyte response and usually fever with a positive RDT.

4. B · Transfuse packed cells in 5–10 mL/kg aliquots aiming for Hb about 7–8 g/dL, reassessing the spleen

B is correct: this is **acute splenic sequestration** with shock. Transfuse urgently but in small aliquots to Hb 7–8 g/dL, because the spleen releases trapped cells as it shrinks (auto-transfusion), and overshooting causes hyperviscosity. A is wrong: Hb 12 g/dL is far above the safe target in SCD. C is wrong: UCG advises avoiding splenectomy in the acute phase (high mortality); it is considered electively after recurrence. D is wrong: repeated crystalloid boluses do not restore oxygen-carrying capacity and delay blood. E is wrong: hydroxyurea is long-term prevention, not crisis treatment.

5. E · Stop the transfusion, keep the IV line open with normal saline and check the patient's identity against the bag

E is correct: fever ≥39 °C, back pain and hypotension are a **category 2 (life-threatening)** reaction, likely acute haemolysis (ABO mismatch) or bacterial contamination. Stop immediately, never restart, keep the line with saline, check identity, resuscitate (NS 20 mL/kg, oxygen) and send the bag and samples. A is wrong: paracetamol and slowing are for mild febrile reactions without other signs. B is wrong: an antihistamine alone suits mild urticaria. C is wrong: furosemide treats overload (TACO), not hypotensive haemolysis, and the transfusion must stop. D is wrong: continuing incompatible blood could be fatal.

6. C · Oral iron (about 3–6 mg/kg/day elemental), albendazole 200 mg, reduce cow's milk to about 500 mL/day and add iron-rich foods, review in 2–4 weeks

C is correct: excess cow's milk, pica, microcytosis and reactive thrombocytosis point to **iron deficiency**; a stable child is treated with oral iron, deworming and diet change, then reviewed for a response. A is wrong: she is compensated, so transfusion is not indicated (UCG: chronic iron deficiency can be corrected with iron alone). B is wrong: folate does not correct iron deficiency. D is wrong: parenteral iron is rarely needed and is a specialist decision because of anaphylaxis risk. E is wrong: thalassaemia trait is less likely with these clues, and treatment need not wait; electrophoresis is reserved for poor response.

7. B · Take a blood culture and give ceftriaxone 80–100 mg/kg IV within 1 hour

B is correct: children with SCD are functionally asplenic, so fever ≥38.5 °C is treated as **pneumococcal sepsis** until proven otherwise; culture and give parenteral ceftriaxone within an hour. A is wrong: delaying antibiotics can be fatal. C is wrong: oral amoxicillin is inadequate for presumed bacteraemia in a febrile SCD toddler. D is wrong: the RDT is negative; treating malaria does not cover the bacterial risk. E is wrong: Hb is at baseline, so there is no transfusion indication.

8. C · Give oxygen, check glucose, and give a simple transfusion of packed cells aiming for Hb about 10–11 g/dL, then refer urgently for imaging and exchange transfusion

C is correct: acute stroke in SCD needs oxygen (SpO_2 >94%), exclusion of hypoglycaemia, and urgent transfusion; exchange is preferred, but if unavailable a simple transfusion to no more than 10–11 g/dL is given, then referral. A is wrong: aspirin and delay are not SCD stroke management. B is wrong: hydroxyurea is for prevention, not acute stroke. D is wrong: Hb 14 g/dL causes hyperviscosity and worsens ischaemia. E is wrong: mannitol and fluid restriction do not address sickling; adequate hydration is needed.

9. A · Packed cells or whole blood 10 mL/kg (130 mL) over 3 hours, no routine furosemide, and do not repeat within 4 days

A is correct: in SAM the heart and kidneys handle sodium and fluid poorly, so WHO limits transfusion to **10 mL/kg over 3 h**, without routine diuretics, and no repeat within 4 days. B is wrong: 20 mL/kg risks fatal heart failure in SAM. C is wrong: the TRACT volume does not apply to SAM. D is wrong: Hb <4 g/dL (or 4–6 with distress) is an indication even in SAM. E is wrong: giving it over 30 minutes is far too fast.

10. D · Yes; hydroxyurea can be offered from 9 months, starting at about 20 mg/kg once daily with FBC monitoring and dose escalation; UCG lists frequent crises as an indication, but international guidance offers it to all children with HbSS

D is correct: NHLBI/ASH guidance and the African trials (REACH, NOHARM) support offering hydroxyurea to all children with HbSS from 9 months, starting around 20 mg/kg/day and escalating with monitoring; UCG 2023 lists specific indications, so explain both. A is wrong: stroke is only one indication. B is wrong: NOHARM showed no increase in malaria. C is wrong: benefit starts in infancy and the start age is 9 months. E is wrong: 35 mg/kg/day is the maximum after escalation, not the starting dose, and starting there risks myelosuppression.

11. C · Hold hydroxyurea, recheck FBC in about 1 week, and restart at a lower dose (e.g. 20 mg/kg) when ANC recovers above 1.0 × 10^9/L

C is correct: ANC below 1.0 × 10^9/L is dose-limiting toxicity; hold, recheck and restart at a lower dose once counts recover (myelosuppression is reversible). A is wrong: escalating with neutropenia worsens toxicity, even though the raised MCV does reflect adherence. B is wrong: permanent stopping is unnecessary and loses benefit. D is wrong: folate does not correct drug-induced neutropenia. E is wrong: G-CSF is not used for hydroxyurea neutropenia and can trigger sickling crises.

12. B · Oxygen, analgesia, incentive spirometry, IV ceftriaxone plus a macrolide, cautious fluids, and transfusion of packed cells

B is correct: fever, chest signs, hypoxia and a new infiltrate define **acute chest syndrome**; management is oxygen, analgesia, incentive spirometry, ceftriaxone plus a macrolide (atypicals), restricted fluids, and transfusion (Hb below 9 g/dL per UCG and falling). A is wrong: hypoxic ACS needs admission. C is wrong: overhydration worsens pulmonary oedema in ACS. D is wrong: infection is a common trigger and antibiotics are essential; opioids also risk hypoventilation. E is wrong: steroids are not routine and are linked to rebound crises.

13. E · Zidovudine (AZT) marrow toxicity

E is correct: **AZT causes macrocytic anaemia** from marrow suppression; Uganda's HIV guidelines advise avoiding or substituting AZT when Hb is low. A is wrong: iron deficiency is microcytic. B is wrong: thalassaemia trait is microcytic. C is wrong: hookworm causes iron-deficiency (microcytic) anaemia. D is wrong: sequestration occurs in SCD with an enlarging spleen and normocytic, high-retic anaemia.

14. A · Blood should be started within 30 minutes of removal from the fridge and completed within 4 hours

A is correct: UCG 2023 states transfusion must start within 30 minutes of removal from the fridge and finish within 4 hours. B is wrong: improvised warming is unnecessary and can haemolyse red cells. C is wrong: routine pre-transfusion diuretics are not needed; give them for heart failure or renal disease. D is wrong: whole blood and red cell concentrates are available from HC IV level. E is wrong: vitals are needed before, at 15 minutes and at the end, at minimum.

15. D · Insecticide-treated net, post-discharge malaria chemoprevention where available, oral iron once recovered, nutrition counselling, and review at 2 weeks and 1–3 months

D is correct: children recovering from severe anaemia have high post-discharge mortality and readmission; ITNs, WHO-recommended post-discharge malaria chemoprevention in high-transmission areas, iron after recovery, feeding advice and early review address the main causes. A is wrong: routine repeat transfusions are not indicated. B is wrong: the post-discharge period is high risk. C is wrong: hydroxyurea is for SCD, which has not been shown here. E is wrong: chloroquine is ineffective against local *P. falciparum* and is not recommended for prophylaxis.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Breathless and pale: severe malarial anaemia

Kato, a 22-month-old boy, is carried into Mengo Hospital paediatric emergency by his mother from Mukono. He has had fever for 4 days and since this morning has been breathing fast and refusing to feed. He was given 'malaria tablets' at a drug shop 2 days ago.

1. What do you do in the first 5 minutes?

> ETAT triage: he is breathing fast and deep with severe palmar pallor (emergency/priority signs). **ABC**: SpO_2 91% → **oxygen**. **IV access**, then draw blood for **HemoCue Hb, malaria RDT, glucose, group and cross-match** before anything is given. Check **CRT** (3 s) and conscious level (alert but irritable: AVPU = A). Avoid a fluid bolus. Request blood urgently.

2. Results: Hb 3.4 g/dL, RDT positive, glucose 4.8 mmol/L. Weight 10.2 kg. HR 172, RR 58 with deep breathing, clear chest, gallop rhythm, liver 4 cm. What is your diagnosis and transfusion decision?

> **Severe malaria with severe anaemia, respiratory distress (acidosis) and heart failure.** Hb ≤4 g/dL is an absolute indication. Give **packed cells 10 mL/kg ≈ 100 mL over 3–4 h** (or whole blood 20 mL/kg ≈ 200 mL if packed cells unavailable) with **furosemide 1 mg/kg (10 mg) IV at the start** because of heart failure. Observe vitals before, at 15 min, every 30 min and at the end.

3. What other treatment starts at the same time?

> **IV artesunate 3 mg/kg** (children <20 kg) at 0, 12 and 24 h then daily until able to take oral, then a full course of artemether-lumefantrine. Paracetamol for fever. Check glucose regularly. Consider antibiotics if sepsis cannot be excluded (ceftriaxone 80–100 mg/kg).

4. Ten minutes after the transfusion starts, he develops urticaria on the trunk. Vitals unchanged, no wheeze. What do you do?

> **Category 1 (mild allergic) reaction**: pause the transfusion, check vitals and ID, give an antihistamine (e.g. **chlorphenamine 0.1 mg/kg** or cetirizine; avoid promethazine under 2 years). If stable and the rash settles, **restart slowly** with close monitoring. If no improvement in 30 min or worsening, treat as category 2.

5. Next morning: Hb 7.6 g/dL, breathing comfortably, liver 2 cm. What is the plan before discharge?

> Complete artesunate → oral ACT. Ask about diet, cow's milk, worms; consider **FBC indices/ferritin**. Start **oral iron** once the acute illness has resolved (≈3–6 mg/kg/day elemental) for 3 months with **albendazole 200 mg** (under 2 years). **ITN**, post-discharge malaria chemoprevention where available, nutrition counselling, check **immunisation card** and **HIV status**, and review at 2 weeks and 1–3 months.

6. How do you present him on the ward round?

> "Kato, 22 months, 10.2 kg (WAZ −1.0), immunised for age, breastfed plus family foods, HIV-negative mother, admitted with severe malaria complicated by severe anaemia (Hb 3.4) with acidotic breathing and heart failure. Received IV artesunate and 100 mL packed cells with furosemide; had a mild urticarial reaction treated with an antihistamine. Now Hb 7.6, stable. Plan: complete ACT, iron after recovery, deworm, ITN, follow-up."

Take-away: Severe anaemia with deep breathing and heart failure is an emergency for blood, not fluids. Transfuse the right volume at the right speed with furosemide if in heart failure, treat malaria at the same

time, and plan follow-up because the risk continues after discharge.

Case 2: The SCD toddler with a growing spleen

Nakato, an 18-month-old girl with HbSS known to the SCD clinic, is brought at 2 a.m. because she is 'too weak to sit' and her tummy looks big. Her mother says she has a mild cold. She is on penicillin V and folic acid.

1. What will you look for on rapid assessment?

ABC; **signs of shock** (HR, CRT, cold peripheries, BP, conscious level); **pallor**; **spleen size** compared with the clinic card; temperature; SpO_2; jaundice; chest signs (ACS). Measure weight (9 kg).

2. Findings: HR 180, CRT 4 s, lethargic, very pale, spleen 9 cm (was 3 cm at clinic), temp 37.9 °C. Hb 2.9 g/dL, retics 18%. What is the diagnosis?

Acute splenic sequestration crisis with shock: rapidly enlarging spleen, steep Hb drop from baseline (≈7 g/dL), high reticulocytes. This can kill within hours.

3. How will you manage her immediately?

Oxygen; IV access; **urgent packed cells in 5–10 mL/kg aliquots** (45–90 mL), reassessing HR, CRT and spleen size; **aim Hb 7–8 g/dL**. If shocked while waiting for blood, a cautious 10 mL/kg crystalloid bolus can be given, but blood is the treatment. Blood culture and **ceftriaxone 80–100 mg/kg** for possible infection; malaria test. Mark the spleen edge with a pen.

4. Why not transfuse 20 mL/kg whole blood to get her Hb to 11?

As the spleen releases sequestered red cells, Hb rises further (auto-transfusion). Large volumes can overshoot to dangerous **hyperviscosity** (stroke, ACS) and overload. Small aliquots with a modest target are safer.

5. She stabilises (Hb 7.4, spleen 4 cm). What long-term plan will you discuss with the family?

Teach parents to **palpate the spleen** daily during illness and come immediately if it enlarges; continue **penicillin V 125 mg 12-hourly**, **folic acid 5 mg**, **monthly SP** malaria prevention, ITN; ensure PCV and pentavalent are complete, PPSV23 at 2 years; **start hydroxyurea** (≥9 months, 20 mg/kg/day); discuss **elective splenectomy** if recurrent or life-threatening episodes (after pneumococcal vaccination), and never in the acute phase.

6. What genetic counselling will you give?

Both parents are likely carriers (HbAS): each pregnancy has a **1 in 4** chance of SCD, 1 in 2 of trait, 1 in 4 unaffected. Offer **testing for siblings** and partner testing for future pregnancies.

Take-away: In SCD, always compare spleen size and Hb with the clinic baseline. Sequestration needs urgent but measured transfusion (5–10 mL/kg aliquots to Hb 7–8) and parent education to catch the next episode early.

Case 3: The pale child who eats soil

Mugisha, 2 years 3 months, is brought to the paediatric outpatient clinic at a HC IV in Wakiso because his grandmother says he is 'pale and lazy'. He is alert and playful.

1. What focused history do you take?

> Diet (breastfed to 6 months? **cow's milk volume**, meat, beans, greens, small fish), **pica** (eats soil), worms, barefoot play, blood in stool or urine, fever and malaria episodes, bleeding or bruising, bone pain or swollen hands (SCD), drugs, **HIV status** of mother and child, immunisation and growth (child health card), development (talking, walking).

2. He drinks about 1.5 L of cow's milk daily, eats little food, plays barefoot and eats soil. Examination: palmar pallor, no jaundice, no splenomegaly, HR 120, RR 28, weight 11 kg (WAZ −1.4). What bedside tests do you do?

> **HemoCue Hb** (6.8 g/dL), **malaria RDT** (negative), and if available an **HbS point-of-care test** (negative). Request FBC with indices, and stool microscopy.

3. FBC: Hb 6.7, MCV 58 fL, MCH 17 pg, platelets 540 × 10^9/L. Stool: hookworm ova. What is your diagnosis and does he need blood?

> **Iron-deficiency anaemia** from dietary deficiency (excess cow's milk) plus **hookworm**. He is clinically **compensated** (no distress, shock or heart failure) and Hb is above 6, so **no transfusion**.

4. Write the treatment.

> **Ferrous sulphate**: UCG 5 mg/kg of salt 8-hourly (≈55 mg 8-hourly; use syrup) giving ≈3–6 mg/kg/day elemental iron, until Hb normal then **3 more months**. **Albendazole 400 mg** single dose (≥2 years), repeat 6-monthly. Diet: **cow's milk ≤500 mL/day**, iron-rich foods (beans, liver, *mukene*, greens), vitamin C, no tea with meals; shoes. Warn about black stools and keep iron out of reach (overdose is dangerous).

5. At review after 4 weeks his Hb is 7.0 g/dL. What do you think about?

> **Poor response**: check adherence (stool colour, pill count), dose, ongoing blood loss (re-check stool), malabsorption, intercurrent infection, wrong diagnosis (**thalassaemia trait**: Mentzer index, electrophoresis; anaemia of chronic disease: HIV/TB), or lead. Check ferritin and retics; refer if no cause.

6. How do you summarise the case for your senior?

> "A 2-year-old, 11 kg (WAZ −1.4), immunised for age, HIV-unexposed, on excessive cow's milk with pica, has microcytic anaemia (Hb 6.7, MCV 58) with reactive thrombocytosis and hookworm. Diagnosis: iron-deficiency anaemia, compensated. Treated with oral iron, albendazole and diet change; review in 2–4 weeks for response."

> **Take-away:** Most chronic anaemia in toddlers is iron deficiency from diet (often too much cow's milk) and worms. A compensated child needs iron, deworming and diet change, not blood, and a check that the Hb actually rises.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.